

HACKATOPIA 2K26 · FINAL PRESENTATION

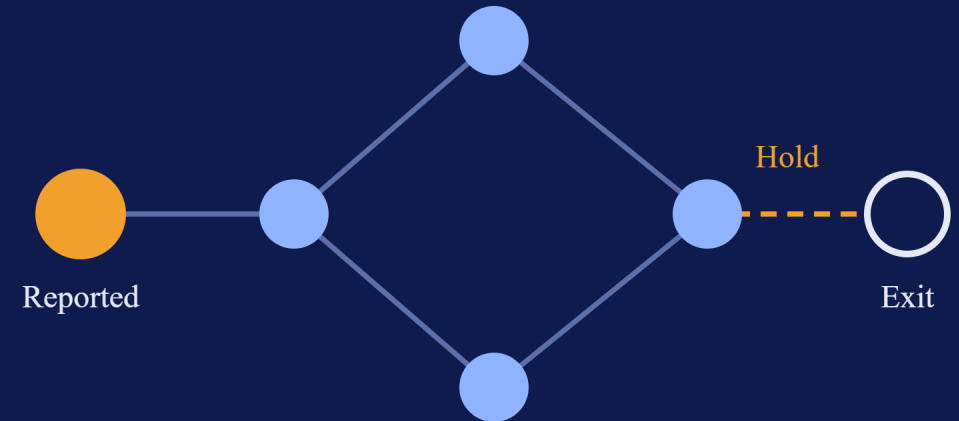
AEGIS-Flow

Counterfactual Temporal Interdiction for
Mule-Account Money Trails

FT-03 · Mule Account Money-Trail Hunter

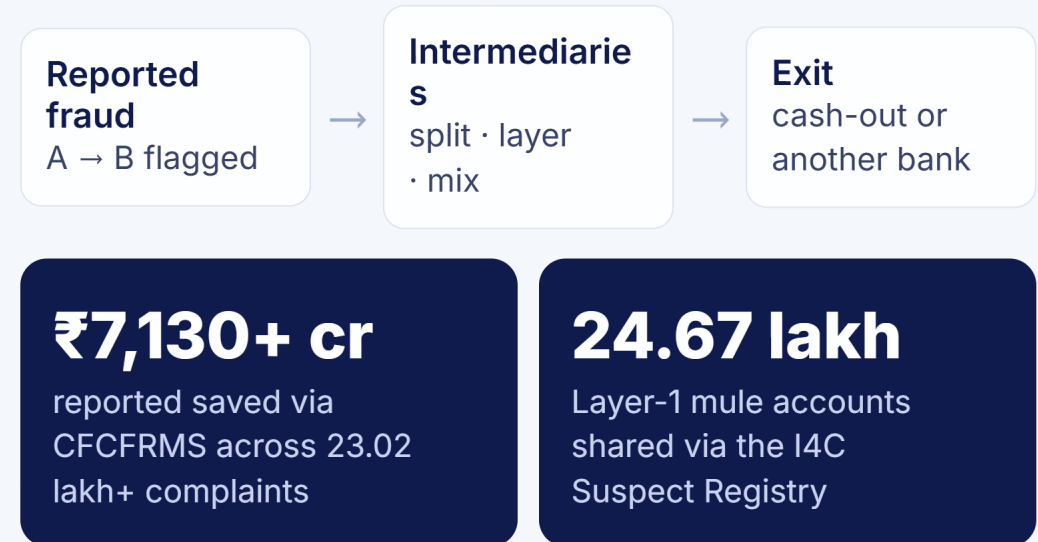
Trace the money. Understand the risk. Evaluate where
to intervene.

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When fraud is reported, the money is already moving.

- **The problem:** reported fraud money is split, passed through mule accounts and mixed with legitimate funds, often across banks.
- **Who is affected:** victims, investigators who rebuild trails by hand, and account holders caught by blunt freezes.
- **Why it matters:** every hop makes recovery harder, and a suspicious-account flag alone does not show where the money went or where to act.
- **Today:** fraud enters through a victim report, a bank's own detection or authorized intelligence; once $A \rightarrow B$ is flagged, we trace $B \rightarrow C \rightarrow D$ and beyond.



Source: Ministry of Home Affairs, parliamentary answer, 9 Dec 2025. National initiatives, not AEGIS-Flow results.

Detection matters. So does understanding the money trail.

Already in place

- Bank fraud monitoring and rule-based alerts
- AI/ML mule detection, such as RBIH's MuleHunter
- I4C: CFCFRMS reporting and the Suspect Registry
- RBIH Digital Payments Intelligence Platform for near-real-time intelligence sharing

What one reported payment still needs

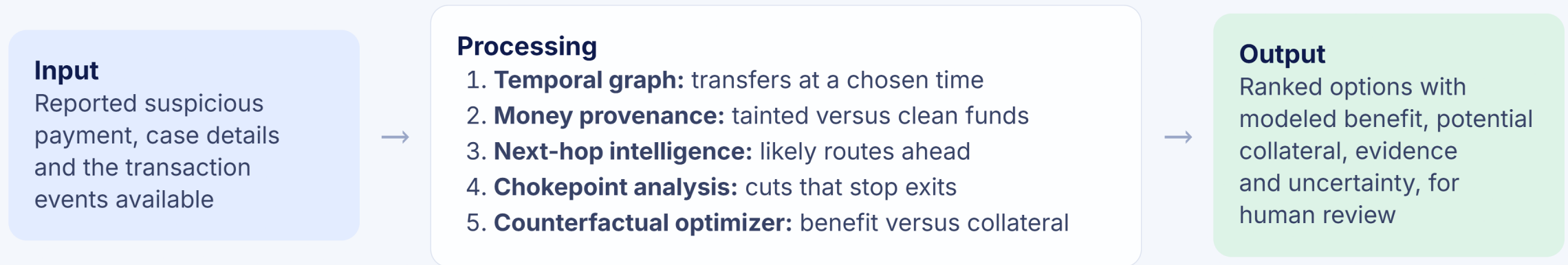
Investigator task	AEGIS-Flow focus
Review a fraud alert	Case-level transaction network
Follow the money	Modeled fund provenance
Consider what comes next	Possible next-hop routes
Compare actions	Simulated intervention trade-offs

Detection says which accounts look suspicious, not where one payment went or what a hold would cost.

The gap: one case-level view that links time-aware reconstruction, tainted-versus-legitimate flows, likely next hops and the cost of each hold. AEGIS-Flow brings these into one prototype workflow and complements existing systems; it does not replace them.

One workspace: reported payment to intervention options.

An investigator-facing system that traces reported fraud money through interconnected transactions and compares where to intervene. It pairs deterministic flow analysis with prediction and counterfactual simulation, and recommends options; people decide.



How it solves the problem: it traces the money rather than the account, separates what was known then from what came later, and shows the cost of each hold before anyone acts.

From flagging accounts to evaluating interventions.

1. **Time-aware investigation:** move the decision time to see what was known then versus observed later.
2. **Money-focused provenance:** models how much reported money moves through splits and mixing, not whole balances.
3. **Counterfactual intervention:** simulates candidate holds, comparing tainted funds intercepted with legitimate funds affected.
4. **Chokepoint identification:** temporal min-cut finds transfers that disconnect a source from an exit.
5. **Explainable uncertainty:** observed, predicted and unavailable information stay visibly distinct.

Every hold, compared

Modeled tainted funds intercepted



Legitimate funds potentially affected



Illustrative; the workspace computes this for each candidate hold.

Unique value: a time-aware workspace for tracing reported money and comparing where intervention may help, with legitimate-fund impact in plain view. The techniques are established; the value is how they work together in one case-level workflow.

From the first click to an evidence-backed recommendation.

1 Case ingestion

Validated multi-bank events become time-ordered contracts.

2 Temporal graph

A decision-time view separates known events from later ones.

3 Money provenance

Tainted and clean funds tracked with zero-loss rounding.

4 Risk and next hops

Model-based risk and ranked likely destinations.

5 Forecast simulation

Bounded future trajectories before funds reach cash-out.

6 Chokepoint search

Capacity-constrained cuts that disconnect source from exits.

7 Counterfactual and Pareto

Intercepted taint weighed against legitimate collateral.

8 Investigator review

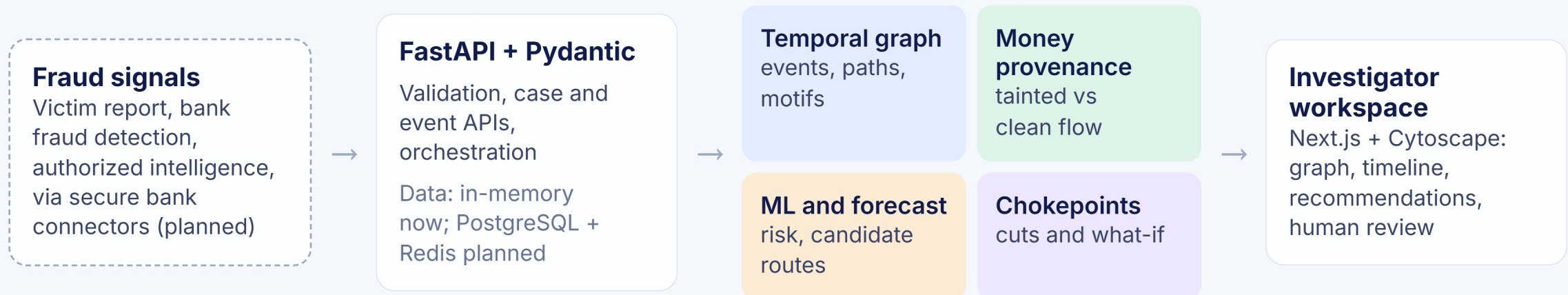
Graph, evidence dossier and diagnostics for human review.

Decision points: pick the decision time T , inspect the evidence, choose an intervention option, then a human approves. **Demo:** slide T to 08:20 UTC and 3 of 7 transfers are available; 4 are later observations.

Proven tools, chosen for typed contracts and fast graphs.

Layer	Technologies	Why we chose it
Frontend	Next.js, React, TypeScript, Tailwind CSS, Cytoscape.js	Typed UI; Cytoscape renders the interactive account graph
Backend and APIs	Python, FastAPI, Uvicorn, Pydantic and Settings, python-dotenv	Validated contracts; REST endpoints for cases, events, graph, timelines, analysis
Graph and core logic	rustworkx, NumPy, Pandas; in-house provenance, min-cut and counterfactual engines	Rust-backed graph traversal; integer minor-unit money math
AI / ML	LightGBM, scikit-learn	Fast tabular risk scoring and next-hop ranking; deterministic analysis runs without the model
Database and cache	PostgreSQL via SQLAlchemy and asyncpg; Redis	Durable case store and fast cache; prototype is in-memory, persistence is next
Cloud and integration	REST today; secure bank connectors, NPCI, I4C, RBIH DPIP planned	Banks keep their data; no live bank feed yet, access needs authorization

Modular services turn transaction events into evidence.



Orchestrator: returns typed results and flags each stage as available, unavailable or insufficient evidence, so partial results stay honest.

Built, integrated and tested, with clear boundaries.

Approach: contract-first and module by module: contracts and APIs, temporal graph, provenance, prediction, chokepoints and counterfactuals, then orchestrator and dashboard.

Challenge and mitigation

Challenge	How we handle it
Future-data leakage	Decision-time boundaries split known from later events
Commingling	Tainted and clean funds modeled separately
Many branches	Traversal, path analysis, candidate cut sets
Model or stage failure	Shows "unavailable"; never fabricates results
Rounding	Integer minor units, largest-remainder allocation

Prototype status

Pipeline and dashboard integrated. 342 backend and 22 frontend tests passed at the last verified checkpoint; lint, TypeScript and production build clean.

Synthetic scenarios: branching, commingling, cold-start, cross-bank, smurfing, rapid pass-through, benign merchant.

Not yet: live bank feed, production database, automated freezes.

From synthetic prototype to governed platform.

Near term

- UX refinement and end-to-end testing
- Packaged, validated model artifacts
- Better calibration and forecast evaluation
- Provenance accounting checked against ground truth

Integration and deployment

- Approved bank-event adapters via secure connectors
- Authenticated ingestion, durable persistence
- Access control, audit trails, privacy safeguards
- NPCI, I4C and RBIH DPIP links, subject to authorization

Expansion

- More payment rails and institutions
- More investigator teams and regions
- Regional-language interface
- Controlled pilots with measured latency and outcomes

Long-term vision: a governed, case-level workspace where authorized investigators reconstruct the money trail, see evidence and uncertainty, and compare interventions across participating sources. These are proposed steps, not current production claims.

From a reported payment to a better-informed decision.

1

Traceability

Follow a reported payment through branching and converging paths in a time-aware graph.

2

Financial context

Model how reported funds move while accounting for legitimate money mixed in.

3

Intervention insight

Compare chokepoints and holds by simulated impact, not a suspicion score alone.

Key takeaway: AEGIS-Flow shows investigators where reported money moved, what the evidence supports and where intervention may help, with uncertainty and legitimate-fund impact in plain view.

Our vision: move fraud investigation from spotting suspicious accounts to understanding the movement of money and where timely action can make a difference.